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6. *(a)* A load of 0.150 kg is attached to the lower end of a light vertical spring of stiffness (spring constant) 7.50 N m^{-1}. The system is initially in equilibrium. A student raises the load vertically through a distance of 0.095 m and releases it so that it oscillates.

Determine:

(i) the extension of the spring when the system is in equilibrium; [2]

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(ii) the period of oscillation. [2]

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(b) The student notices the amplitude of oscillation decreasing and records the following amplitudes.

Oscillation number (n)	Amplitude (A) /m
0	0.095
10	0.062
20	0.043
30	0.029
40	0.019
50	0.014
60	0.009

She comments that "the rate of decrease of amplitude is larger at the start of the experiment than at its end". Justify the statement using the data from the table. [2]

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(c) She expects that the measured amplitude of oscillation can be described by the equation:

$$A = A_0 e^{-\frac{n}{N}}$$

where A_0 is the initial amplitude, n is the oscillation number and N is a constant for the experiment that describes the decay of the amplitude.

(i) Justify that A_0 is the amplitude when n = 0. [1]

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(ii) By considering the amplitude when n = 30 use the equation to determine a value for N. [2]

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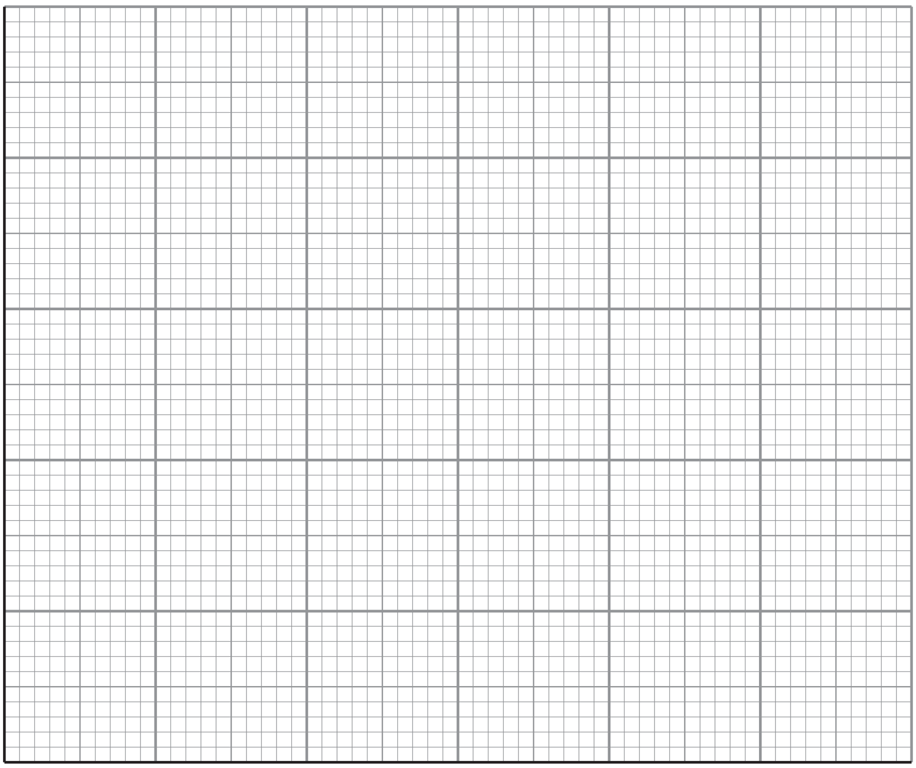
(d) The student decides to check the validity of the equation.

(i) Complete the third column in the table. The first four rows have already been completed. [1]

Oscillation number *(n)*	Amplitude *(A)* / m	$\ln\left(\frac{A_0}{A}\right)$
0	0.095	0
10	0.062	0.43
20	0.043	0.79
30	0.029	1.19
40	0.019	
50	0.014	
60	0.009	

(ii) Plot $\ln\left(\frac{A_0}{A}\right)$ (y-axis) as a function of the oscillation number (x-axis) and draw a line of best fit. [3]



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(iii) Explain whether or not your graph is in agreement with the equation $A = A_0 e^{-\frac{n}{N}}$. [2]

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(iv) Use the graph to determine a value for N. [3]

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(v) Explain which of the two values obtained for N is expected to be the more accurate, the value in part *(c)*(ii) or *(d)*(iv). [1]

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